MIL-T-82607B(OS)
17 November 1976
Superseding
MIL-T-82607A(OS)
Amendment 1
23 August 1974

MILITARY SPECIFICATION

TELEVISION SYSTEM MK 5 MODS 0 AND 1

* This specification is approved for use by the Naval Sea Systems Command (OS), Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers the requirement for a closed circuit television system which functions primarily to provide long range electro-visual target tracking for shipboard gun and guided missile fire control systems under conditions of both normal (daylight) and very low light levels.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-17/7	Cable, Radio Frequency, Coaxial, RG12A/U (NWR-8)
MIL-C-17/36	Cable, Radio Frequency, Coaxial, RG79B/U
MIL-P-116	Preservation - Packaging, Methods of
MIL-S-901	Shock Tests, H. I. (High-Impact), Shipboard Machinery, Equipment and Systems, Requirements for
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-F-18870	Fire Control Equipment, Naval Shipboard, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Division (611), Indian Head, MD 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-T-28800	Test Equipment for use with Electrical and Electronic Equipment, General Specification for
MIL-H-46855	Human Engineering Requirements for Military Systems, Equipment and Facilities

STANDARDS

Federal

FED-STD-151	Metal, Test Methods

Military

MIL-STD-108	Definition of and Basic Requirement for Enclosure for Electric and Electronic Equipment
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment
MIL-STD-210	Climatic Extremes for Military Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurements of
MIL-STD-470	Maintainability Program Requirements (for Systems and Equipments)
MIL-STD-740	Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-831	Test Reports, Preparation of
MIL-STD-1399, SECTION 103	Interface Standard for Shipboard Systems, Electric Power, Alternating Current

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MS 17346	Connectors, Receptacle, Electrical, Box Mounting
MS 3102	Connectors, Receptacle, Electric, Box Mounting, Solder Contacts, AN Type
MS 3108	Connectors, Plug, Electric, Solder Contact, 90 Degree, AN Type

Industrial

EIA RS 170	Monochrome Television Studio Facilities
IRE Standard 1954	Method of Measurement of Aspect Ratio & Geometric Distortion

DRAWINGS

Naval Sea Systems Command

2807052	Director MK 73 MODS 1 and 2 CAMERA SUPPORT
2928175	Maintenance Kit
IL-2928400	Television System MK 5 MOD 0
2835715	Control Amplifier MK 163 MOD 3
2537504	Optical Sighting System MK 1 MOD 2
2928101-2	Television Camera MK 15 MOD 1
2993658	Gun Control Console MK 68 MOD 4

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 System Definition. - The Television System MK 5 is designed to provide an electro-optical display for use in target tracking for shipboard gun and guided missile fire control systems under normal conditions of daylight and night time operations. The Television System must be able to operate in a high intensity field such as may be produced by high level energy radiated by a fire control radar, search radar, or communications transmitter.

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3.1.1 Low Light Level Television System MK 5 MOD 0. - The Low Light Level Television System (LLLTVS) MK 5 MOD 0 is comprised of one (1) Television Camera MK 15 MOD 0 per Dwg. 2928101G1, one (1) Television Monitor MK 30 MOD 0 per Dwg. 2928102G1, and one (1) Maintenance kit MK 86 MOD 0 per Dwg. 2928175G2 or MOD 1 per Dwg. 2928175G3.

3.1.2 Low Light Level Television System MK 5 MOD 1. - The Low Light Level Television System MK 5 MOD 1 is comprised of one (1) Television Camera MK 15 MOD 1 per Dwg. 2928101G2, and one (1) Maintenance kit MK 86 MOD 2 per Dwg. 29281754G4. The Television Monitor MK 30 MOD 0 will also be a part of the LLLTVS MK 5 MOD 1 if specified by the contract.

3.1.3 Interface Definition - LLLTVS MK 5 MOD 0. - When installed in the Fire Control SYstem MK 74 the TV System will interface with the Director, the radar console, and the missile fire control switchboard physically and electrically as indicated by Figures 1, 2 and 3. Power (115 V, 400 Hertz) to the system will be supplied from the Missile Fire Control Switchboard to the ON/OFF switch on the radar console. From the radar console, power is applied through the radar slip rings and left elevation junction box to the camera where the 115V, 400 Hertz is converted to 24 VDC $^{+\ 4V}_{-\ 1V}$ in the camera primary power supply. The 24 VDC is then used as the power supply for the camera electronics, relay controls, and switch controls. Heater power (115 volt 60 or 400 Hertz) is obtained (separate from system power) from the Missile Fire Control Switchboard. (See Addendums A and B)

3.1.4 Interface Definition - LLLTVS MK 5 MOD 1. - When installed in the Fire Control System MK 86, the TV Camera MK 15 MOD 1 will interface with the Optical Sighting System MK 1 MOD 2 (OSS) the Gun Control Console MK 68 MOD 4 (GCC) and the Gun Fire Control Switchboard as indicated by Figures 4 and 5. Power (22 VDC ± 1V) to the system will be supplied form the OSS box to the camera and relays controlled by the TV System ON/OFF switch on the GCC. Heater power (115V, 400 Hz), which is separate from system power, is obtained and continuously supplied from the Fire Control Switchboard. Synchronizing pulses are supplied from the Video Amplifier Synchronizer MK 161.

3.1.5 General Description of the Basic Equipments

3.1.5.1 Camera Group. - The camera group shall conform to the requirements as specified herein and shall consist of a lens system, fiber optic coupled intensifier, a sensor tube, and associated electronics. The MK 15 MOD 0 camera will be mounted on the camera platform of the Director MK 73. The MK 15 MOD 1 camera will be mounted on the camera platform of the OSS.

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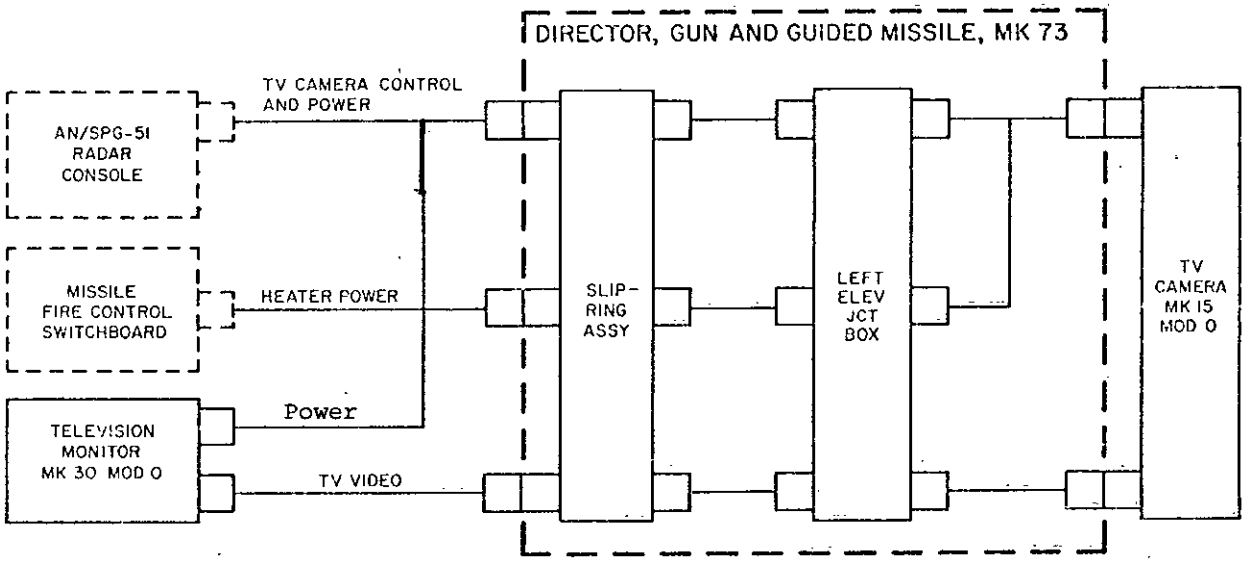


FIGURE 1

TYPICAL BLOCK DIAGRAM

TELEVISION SYSTEM MK 5 MOD 0

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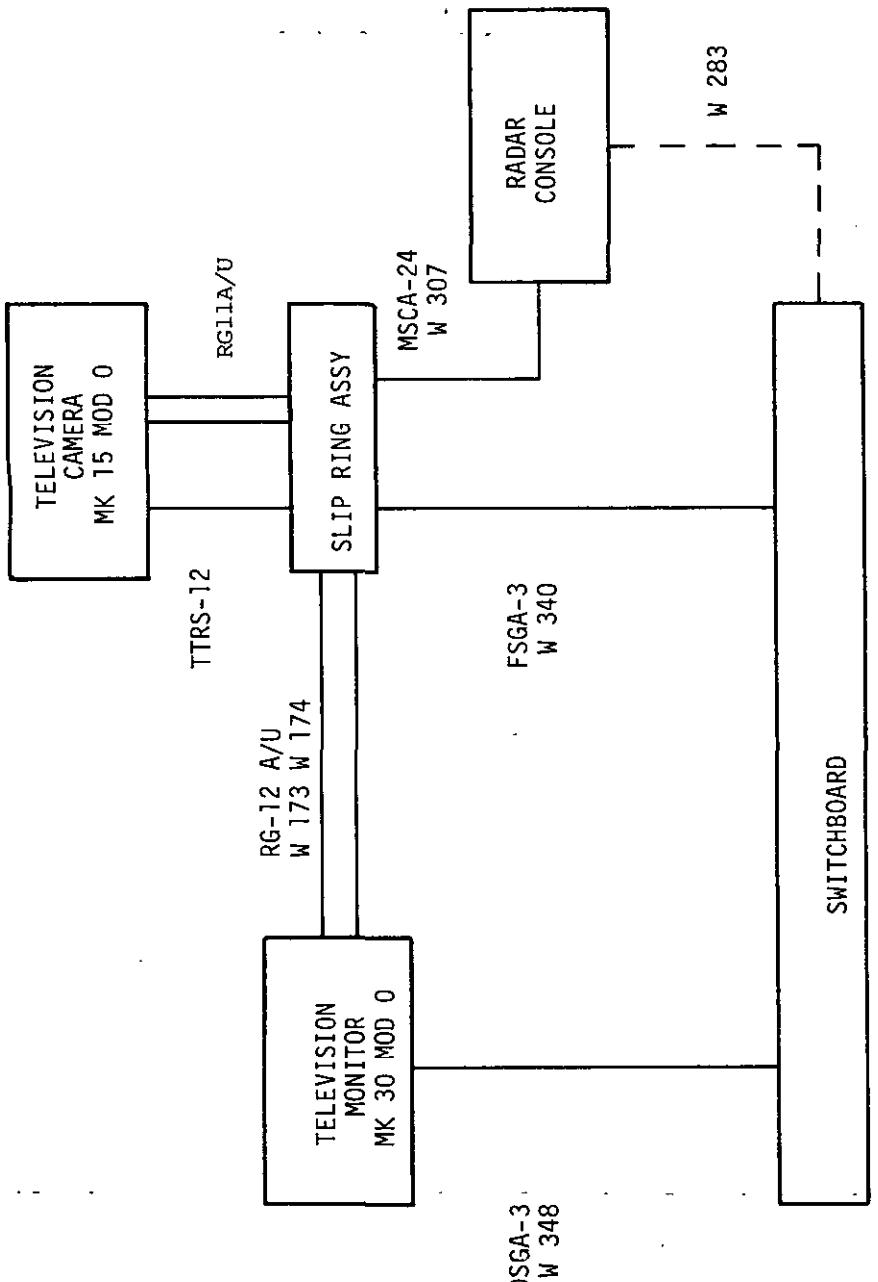


FIGURE 2
CABLING DIAGRAM
TELEVISION SYSTEM MK 5 MOD 0

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TV CAMERA MK 15 MOD 0

P - A: 115V 400 Hz ϕA
P - B: 115V 400 Hz ϕA Return
P - C: 115V 60 Hz ϕA Heater
P - E: 115V 60 Hz ϕA Heater Return
- J: Signal Ground
- S: Filter Select - 1
- T: Filter Select - 2
- U: Filter Select - 3
- V: Filter Select - 4
- W: Filter Select - 5
- H: Filter Select - 6
- X: Filter Position Indicator
- Y: Field of View Select
P - b: Video Inversion Control Return
P - d: Video Inversion Control
- D: Depth of Field Select
- K: Spare
- L: Spare
F: Spare
G: Spare
- M: Spare
- N: Spare
- P: Spare
- R: Spare
- Z: Spare
- a: Spare

TV MONITOR MK 30 MOD 0

Connector Type MS-3108R18-4S

P - A: 115V 400 Hz ϕA

P - B: 115V 400 Hz ϕA Return

(P = Shielded pair)

Connector Type UG-603B/U

P2 and P1 - VIDEO (coax)

FIGURE 3

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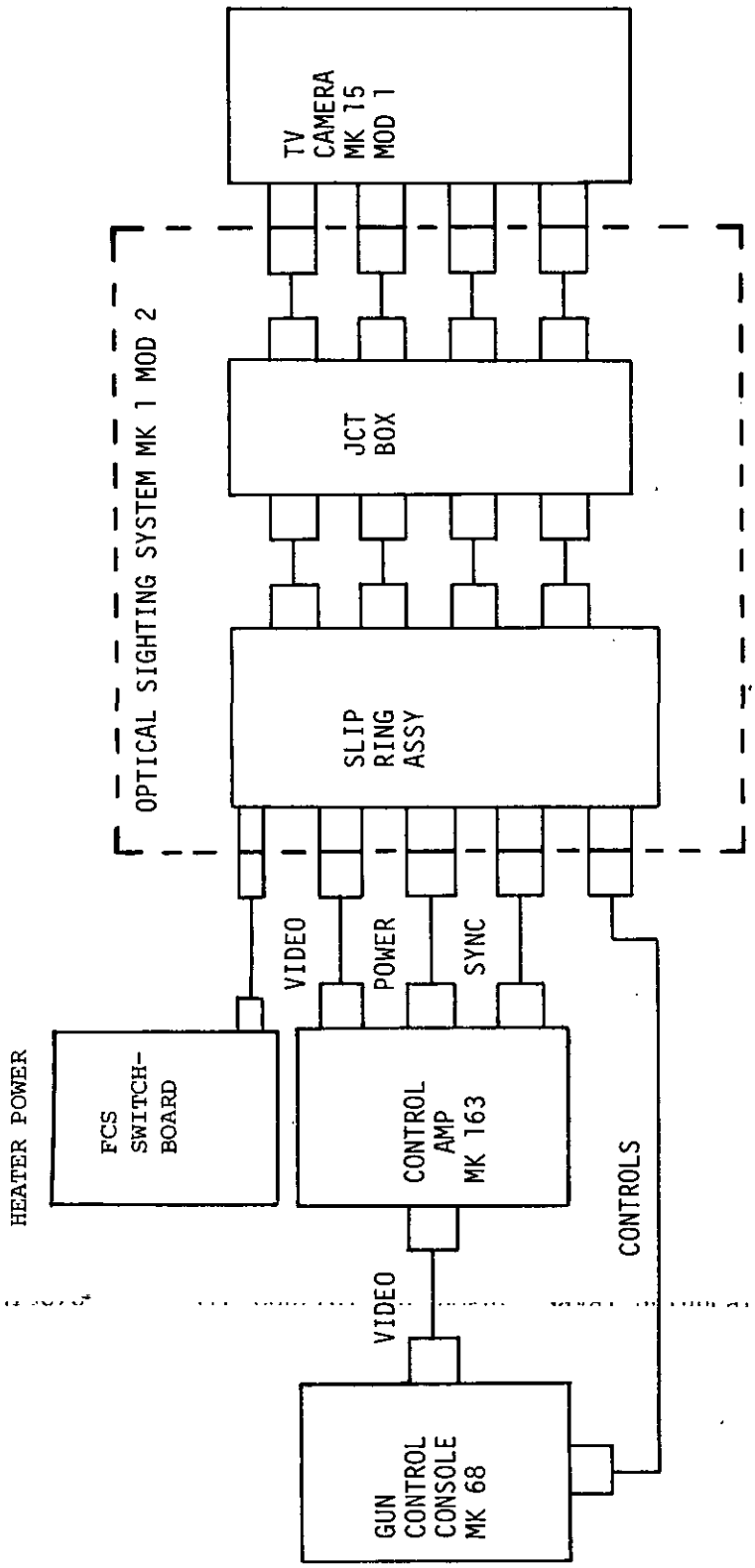


FIGURE 4
INTERFACE DIAGRAM
TV CAMERA MK 15 MOD 1

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TV CAMERA MK 15 MOD 1

CONTROL/POWER CONNECTOR J1

-B	+22 VDC
P -C	115V, 400 Hz, ϕA
P -E	115V, 400 Hz, ϕA Return
-F	Gray Scale
-J	+22 VDC Return
-S	Filter Select 1 or 4
-T	Filter Select 2 or 5
-U	Filter Select 3 or 6
-V	Iris Control
-W	+26 VDC Return
-X	Video Invert
-Y	Field of View Select
-b	Day/Night Filter Select
-A	Spare
-D	Spare
-H	Spare
-K	Spare
-L	Spare
-M	Spare
-N	Spare
-P	Spare
-R	Spare
-Z	Spare
-a	Spare

FIGURE 5

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-d	Spare

Coax Connectors

-P1	(+) Video
-P2	(-) Video
-P3	Field Drive
-P4	Line Drive

FIGURE 5 (Continued)

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3.1.5.1.1 Camera Group Enclosure - The camera group shall consist of three subassemblies - lens assembly, camera head, and camera electronics. The lens assembly shall have two compartments, a watertight electronic section and an airtight optical section. The camera head assembly shall be airtight when mounted to the optics assembly. The camera assembly shall be watertight when mounted to the optics assembly.

3.1.5.1.2 Lens Assembly. - The lens assembly shall have two fixed focal length lenses having a horizontal field of view (FOV) of 1 and 5 degrees respectively as specifically defined in 3.2.4.11.1 of this document. The lens assembly shall be mounted in the air and watertight enclosure specified in 3.2.4.1 along with a protective shutter subassembly, a filter subassembly, a FOV selection subassembly, a heater subassembly, and a high light level sensing subassembly. The filter and FOV subassemblies shall be remotely controlled, while the protective shutter subassembly will be controlled by the light level impinging on the photo-electric cells of the high light level subassembly to prevent damage to the image intensifier/sensor subassemblies. The heater subassembly shall include a thermostatic switch to energize the heater at temperatures below 5°C (41°F) and shall provide heat adequate to prevent icing of the protective shutter or window under environment conditions stated in 3.5.4.

3.1.5.1.3 Camera Head Subassembly. - The camera head subassembly shall be mounted to the lens assembly. The subassembly shall consist of a micro-channel plate image intensifier, fiber-optic coupled to the sensor tube, and the preamplifier, blanking, and associated electronics.

3.1.5.1.4 Camera Group Electronics - TV Camera MK 15 MOD 0. - The Camera Group Electronics shall include those circuits necessary to generate the sensor tube synchronizing and sweep voltages, process video, provide necessary power supplies, and provide target gain control. The power supplies shall be in separate shielded compartments. The electronics shall also include in the optics the control circuitry for operating the protective shutter, changing the filters, and opening the outside window cover. The electronics shall consist of printed circuit cards where practical.

* 3.1.5.1.5 Camera Group Electronics - TV Camera MK 15 MOD 1. - The Camera Group Electronics shall include those circuits necessary to process video, provide necessary power supplies, and provide target gain control. The power supplies shall be in separate shielded compartments. The electronics shall also include in the optics the control circuitry for operating the protective shutter, changing the filters, and opening the outside window cover. The electronics shall consist of printed circuit cards where practical.

3.1.5.2 Controls and Indicators. - Except for monitor and camera operational and maintenance adjustment, all other operational controls for the TV System will be mounted on consoles of the Fire Control Systems.

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3.1.5.2.1 LLLTVS MK 5 MOD 0 Controls. - The controls shall consist of the following:

3.1.5.2.1.1 System ON/OFF Switch.- The system control switch shall have two positions: ON/OFF. In the ON position the TV system will be fully energized and operating. The outside window cover shall be open. In the OFF position the TV System is completely deenergized except for heater power. In the OFF position the outside window cover shall close.

3.1.5.2.1.2 MAGNIFICATION Select. - The magnification select switch shall be a two position switch which will allow the manual selection of either the X1 (5 degree FOV) or the X5 (1 degree FOV).

3.1.5.2.1.3 FILTER Select. - The filter select switch shall be a six position switch which will allow the manual selection of filters for insertion into the optical line-of-sight. The filters shall include: Clear, Red, Vertical Polarized, and Neutral Density (ND) and combinations of the aforementioned filters. The filters shall be mounted on one wheel. Legends for the filter control shall read "NIGHT" with three subheadings "CLEAR", "GLARE", "HAZE" and "DAY" with three subheadings, "TWILIGHT", "DAYLIGHT", and "DIRECT SUN".

3.1.5.2.1.4 Video POSITIVE/NEGATIVE Select. - The video inversion switch shall be a two-position switch which will allow manual selection of either normal (positive) video or inverted (negative) video.

* 3.1.5.2.1.5 CLOSE/NORMAL Switch. - The CLOSE/NORMAL Switch shall be a two position switch which enables the manual selection of either a close or normal depth-of-field. Selection of the CLOSE position shuts the modified front door of the camera. The modified door shall have two apertures, one for each FOV. The apertures, which shall be fitted with optically flat glass, shall result in an f/19 system with the door in the closed position.

3.1.5.2.1.6 Filter Position Indicator. - The filter position indicator shall be a light which will indicate when the filter has reached the desired position.

3.1.5.2.2 LLLTVS MK 5 MOD 1 Controls. - The controls provided on the Gun Control Console MK 68 MOD 4 Consist of the following:

* 3.1.5.2.2.1 Camera ON/OFF and Gray Scale Control. - The ON/OFF control as provided will be a two function rotary switch connected such that in the full counterclockwise (CCW) position the camera is deenergized except for heater power. Full clockwise (CW) position will apply power and present a gray scale for monitor adjustment. Partial CW position applies camera power only.

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* 3.1.5.2.2.2 FOV Select. - The FOV select control will be a three position momentary, center off, switch which will allow selection of either the five degree FOV (wide) or the one degree (narrow).

* 3.1.5.2.2.3 Filter Select. - Filter selection will be provided by using two switches. One switch, a three position momentary, center off, will allow selection of DAY or NIGHT filters. The other switch, a three position rotary, will select one of three DAY filters: DARK, LIGHT or SUN, or one of three NIGHT filters: CLEAR, GLARE, or HAZE.

* 3.1.5.2.2.4 Video Positive/Negative Select. - The video inversion select will be a rotating potentiometer control. Full CCW position will select negative video, full CW position shall provide positive video.

* 3.1.5.2.2.5 Iris Control. - The Iris control, as provided, is a three position, momentary, center off, switch which allows the selection of an OPEN or CLOSE position of the lens door on the MK 15 MOD 1 camera. This door is fitted with two apertures, one for each field of view. The apertures, which shall be fitted with optically flat glass, shall result in an f/19 system with the door in the closed position.

* 3.1.5.2.2.6 Reticle Select. - The electronic reticle select will be a rotating potentiometer control. Full CCW position will display the reticle; full CW position will turn reticle off.

3.1.5.3 Monitor. - The monitor shall be a 14-inch monitor utilizing solid state components in plug-in circuit boards, specifically constructed for shipboard use. The monitor will be mounted in the fire control room above the radar console. This Monitor is a MK 30 Mod 0.

3.1.5.4 Maintenance Kit MK 86. - The Maintenance Kit MK 86 per NAVORD Dwg. 2928175 shall be ordered by MK and MOD number in the quantity stated in the contract or order. The Maintenance Kit is defined as follows:

3.1.5.4.1 Maintenance Kit MK 86 MOD 0. - The Maintenance Kit MK 86 MOD 0 shall consist of one each of the following items, unless otherwise specified.

a. Carrying Case Dwg. No 2928468-1 (Para. 3.6)

b. Nine Ready Service Spares consisting of:

(1)	Control Regulator	Dwg. No. 2928031-1
(2)	Video Processor	Dwg. No. 2928032-1
(3)	Sync Generator	Dwg. No. 2928033-1
(4)	Deflection-Fail/Safe	Dwg. No. 2928034-1

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(5)	Synchronizer	Dwg. No. 2928021-1
(6)	Video Amplifier	Dwg. No. 2928023-1
(7)	Vertical Amplifier	Dwg. No. 2928037-1
(8)	Horizontal Amplifier	Dwg. No. 2928038-1
(9)	Optics Logic Control (Amendment 1 dated 8/23/74)	Dwg. No. 2928752-1

c. 1 ea Cable Assembly Dwg. No. 2928425-1

d. 2 ea Cable Assembly Dwg. No. 2928076-1

e. 1 ea Cable Assembly Dwg. No. 2928469-1

f. 1 ea Spanner Wrench Dwg. No. 2928470-1

g. 1 ea Lens Cap Dwg. No. 2928471-1

h. 4 ea Tip Jacks Dwg. No. 2928475-1

3.1.5.4.2 Maintenance Kit MK 86 MOD 1. - The Maintenance Kit MK 86 MOD 1 shall consist of the following items in the quantity specified unless otherwise specified:

a. 1 ea Carrying Case Dwg. No. 2928468-1 (Para 3.6)

b. Two (2) ea of the following Ready Service Spares:

(1)	Control Regulator	Dwg. No. 2928031-1
(2)	Video Processor	Dwg. No. 2928032-1
(3)	Sync Generator	Dwg. No. 2928033-1
(4)	Deflection - Fail/Safe	Dwg. No. 2928034-1
(5)	Synchronizer	Dwg. No. 2928021-1
(6)	Video Amplifier	Dwg. No. 2928023-1
(7)	Vertical Amplifier	Dwg. No. 2928037-1
(8)	Horizontal Amplifier	Dwg. No. 2928038-1
(9)	Optics Logic Control (Amendment 1 dated 8/23/74)	Dwg. No. 2928752-1

c. 1 ea Cable Assembly Dwg. No. 2928425-1

d. 2 ea. Cable Assembly Dwg. No. 2928076-1

e. 1 ea Cable Assembly Dwg. No. 2928469-1

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f. 1 ea. Spanner Wrench Dwg. No. 2928470-1

g. 1 ea. Lens Cap Dwg. No. 2928471-1

h. 4 ea. Tip Jacks Dwg. No. 2928475-1

3.1.5.4.3 Maintenance Kit MK 86 MOD 2. - The Maintenance Kit MK 86 MOD 2 shall consist of the following items, unless otherwise specified:

a. Carrying Case Dwg. No. 2928504-1

b. Ready Service Spares consisting of:

(1) Control Regulator Dwg. No. 2928031-1

(2) Video Processor Dwg. No. 2928505-1

(3) Sync Generator Dwg. No. 2928506-1

(4) Deflection-Fail/Safe Dwg. No. 2928034-1

(5) Optics Logic Control Amendment 1 dated 8/23/74 Dwg. No. 2928752-1

c. 1 ea. Spanner Wrench Dwg. No. 2928470-1

d. 1 ea. Lens Cap Dwg. No. 2928471-1

e. 4 ea. Tip Jacks Dwg. No. 2928475-1

3.2 Specific Requirements. - The TV System shall function as an integral part of the Fire Control System to provide an electro-optical display for use in tracking targets. Performance of equipments shall be as specified herein.

3.2.1 Primary Power Requirements. - The TV System shall operate from the following sources of power as defined in MIL-STD-1399 Section 103:

a. LLLTVS MK 5, MOD 0

115 Volts, 400 Hertz Single Phase, Type II, 3.5 amperes maximum.

b. Camera MK 15 MOD 0 Heater Power (Continuous):

115 Volts, 60 or 400 Hertz, Single Phase, Type II, 3.5 amperes maximum.

* c. Camera MK 15 MOD 1 Heater Power (Continuous)

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115 Volts, 60 or 400 Hertz, Single Phase, Type II, 3.5 amperes maximum

d. Camera MK 15 MOD 1 Sensor Power:

±26 Volts dc, 0.5 amperes maximum
±22 Volts dc, 2.5 amperes maximum

3.2.1.1 Overload Protection. - Neither fuses nor circuit breakers shall be used in the camera enclosure.

3.2.2 Resolution. - The resolution requirements for the LLLTVS MK 5 MODS 0 and 1 are as follows:

3.2.2.1 Resolution - LLLTVS MK 5 MOD 0. - When viewing a 100% contrast target, illuminated by a 2870°K source, through either the long or short focal length lens, the TV System shall resolve a minimum of 550 TV lines per picture height with a faceplate illumination of 10^{-3} foot candles and a minimum of 250 TV lines per picture height to 30°C and 150 TV lines from 30° to 50°C with a faceplate illumination of 5×10^{-7} foot candles (a 25% degradation will occur from 50°C to 55°C as specified herein).

3.2.2.2 Resolution - LLLTVS MK 5 MOD 1. - When viewing a 100% contrast target, illuminated by a 2870°K source, through either the long or short focal length lens, the TV Camera shall resolve a minimum of 650 TV lines per picture height with a faceplate illumination of 10^{-3} foot candles and a minimum of 300 TV lines per picture height to 30°C and 180 TV lines from 30°C to 50°C with a faceplate illumination of 5×10^{-7} foot candles (a 25% degradation will occur from 50°C to 55°C as specified herein).

3.2.3 Wiring and Cabling.

3.2.3.1 Grounding. - The outside enclosures of all units or assemblies and all external parts shall be at ground potential at all times exclusive of antenna and transmission lines which shall be at ground potential except in regard to potential actually to be transmitted.

3.2.3.2 Cabling. - Coaxial and control and power cables, in accordance with MIL-F-18870, shall be provided to allow correct cabling of the camera and monitor into existing or developing systems and equipments.

3.2.4 Camera Requirements. - The TV Camera MK 15 all MODS shall be one enclosure containing the following functional units:

a. Lens System

b. Head Assembly containing Image Intensifier and Sensor Tube

c. All Camera Power Supplies

d. All Camera Electronics

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3.2.4.1 Enclosure. - The enclosure shall consist of three (3) compartments each housing one of the functional units listed above. The lens system shall be airtight and filled with dry air or nitrogen to "0" PSIG. The head assembly containing the intensifier and sensor tube shall be airtight when mounted to the lens assembly and filled with dry air or nitrogen to "0" PSIG by means of two separate valves, one for intake and the other for exhaust. The electronics compartment shall be watertight and shall have one (1) access cover for maintenance purposes.

* 3.2.4.1.1 Alignment. - The camera shall include mounting adjustments so that the optical axis of the TV System can be aligned parallel to the optical axis of the fire control director boresight telescope to within plus or minus one-half milliradian. Adjustment of ±1° shall be provided for optical alignment in the horizontal and vertical planes. Parallels will be computed by the fire control computer to compensate for cameras physically located away from the fire control director. In all cases the alignment accuracy requirement remains the same.

3.2.4.1.2 Window. - The optically clear window coupling the enclosed lens systems to the atmosphere shall have a remotely operated cover to protect the window from the elements during periods when the camera is not in use. The window may be a refractive element.

3.2.4.2 Resolution. - The resolution requirements for the Camera MK 15 MODS 0 and 1 are as follows:

3.2.4.2.1 Resolution - Camera MK 15 MOD 0. - When viewing a 100% contrast target, illuminated by a 2870°K source, through either the long or short focal length lens, the camera shall resolve a minimum of 550 TV lines per picture height with a faceplate illumination of 10^{-3} foot candles and a minimum of 250 TV lines per picture height to 30°C and 150 TV lines from 30°C to 50°C with a faceplate illumination of 5×10^{-7} foot candles (a 25% degradation will occur from 50°C to 55°C as specified herein).

* 3.2.4.2.2 Resolution - Camera MK 15 MOD 1. - When viewing a 100% contrast target, illuminated by a 2870°K source, through either the long or short focal length lens, the camera shall resolve a minimum of 650 TV lines per picture height with a faceplate illumination of 10^{-3} foot candles and a minimum of 300 TV lines per picture height to 30°C and 180 TV lines from 30°C to 50°C with a faceplate illumination of 5×10^{-7} foot candles (a 25% degradation will occur from 50°C to 55°C as specified herein).

* 3.2.4.3 Field and Frame Requirements. - The Camera MK 15 MOD 0 and 1 shall operate as follows:

	MK 15 MOD 0	MK 15 MOD 1
Frame Rate (frames/sec)	30	22.2
Field Rate (2X frame rate)	60	44.4
Interlace (positive) rates	2:1	2:1
Scan Line Per Frame	525	729
Aspect Ratio	4:3	1:1

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3.2.4.4 Sweep. - Sweep non-linearity (deviation from the best straight line) shall not exceed ± 2% during the active scan time for either the vertical or horizontal applied sweep voltages when measured as specified in IRE Standards on Television: Method of Measurement of Aspect Ratio and Geometric Distortion, 1954, using 5% geometric distortion as the requirement for the entire camera.

3.2.4.5 Automatic Gain Control. - The camera shall have 30 db of automatic gain (level) control (AGC).

3.2.4.6 Automatic Target Control. - The camera shall incorporate automatic target control which shall sense the peak video and in conjunction with AGC shall maintain the camera output within plus or minus 15 per cent over a light level ratio of 10^4.

3.2.4.7 Manual Target Control. - Manual target control shall be provided such that when selected the control shall vary the vidicon target voltage to enhance background contrast. This control shall be internal to the camera.

3.2.4.8 Gray Scale Reproduction. - The camera shall produce a uniform flat picture field and not less than six (6) shades of gray when viewed on an EIA test chart (as specified in IRE Standards on Television, *Method of Measurements* of Aspect Ratio and Geometric Distortion, 1954) with a faceplate illumination of 10^{-3} foot candles.

3.2.4.9 Electronics. - Camera electronics shall include horizontal and vertical deflection circuits, snyc generation circuits, video amplification (pre-amplifier), control circuits and power supplies.

3.2.4.9.1 Power Supplies. - Camera electronics shall operate from self contained electronically regulated dc voltage supplies with input power supplied from primary sources as specified herein.

3.2.4.9.2 Sync Generator. - The sync generator of Camera MK 15 MOD 0 shall provide for full 2:1 positive interlace scanning of the sensor tube and shall provide synchronizing pulses as defined in EIA Standard RS 170. The sync generator of Camera MK 15 MOD 1 shall provide for full 2:1 positive interlace scanning of the sensor tube and shall accept external sync pulses generated by the Control Amplifier MK 163.

3.2.4.9.3 Sweep Failure Protection. - Sweep failure protection circuitry shall be included to prevent damage to the pick-up tube in event of failure of either horizontal or vertical deflection, or both.

3.2.4.9.4 Adjustments. - Internal set-up adjustments shall include Horizontal Size; Horizontal Centering; Vertical Size; Vertical Centering; Focus Current Adjust; Auto Target Threshold Adjust (Light Level).

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3.2.4.9.5 Video Output. - The video outputs from the MK 15 MOD 0 and MOD 1 cameras, when the faceplate is illuminated by 10^{-3} foot candles, shall meet the following requirements.

3.2.4.9.5.1 MK 15 MOD 0 Camera Video Output. - The camera shall produce composite video of 1.3 ± .8 volts peak-to-peak when terminated with an impedance of 75 ohms.

* 3.2.4.9.5.2 MK 15 MOD 1 Camera Video Output. - The camera shall produce non-composite video of 3.0 ± 1.5 volts peak-to-peak on the positive and negative video outputs when terminated with an impedance of 120 ohms. The two video outputs shall be of opposite polarity.

3.2.4.9.6 Output Cable - Camera MK 15 MOD 0. - The Camera MK 15 MOD 0 shall drive 100 feet of coaxial cable having characteristics of RG 12A/U as specified in MIL-C-17/7 and having a termination of 75 ohms.

* 3.2.4.9.7 Output Cable - Camera MK 15 MOD 1. - The camera MK 15 MOD 1 shall drive 300 feet of coaxial cable having characteristics of RG 79B/U as specified in MIL-C-17/36 and having a termination of 120 ohms.

3.2.4.10 Image Intensifier and Sensor. - This system shall comprise a single-stage micro-channel plate intensifier and a pick-up tube, coupled by fiber-optics.

3.2.4.11 Lens System. - The lens system shall be comprised of two fixed focal length lens assemblies having horizontal FOVs of 1 and 5 degrees respectively as defined by the focal lengths specified below. The lens assemblies shall be fixed-mounted in an airtight enclosure.

3.2.4.11.1 Lens Requirements. - The lens assembly shall be as follows:

NORMAL Depth-of-Field:

Focal Length	8 inches (±2%)	40 inches (±2%)
Speed	T/3.3 (or faster)	T/6 (or faster)
Depth-of-Field	500 yds to 50,000 yds	5550 yds to 50,000 yds

CLOSE Depth-of-Field:

Focal Length	8 inches (±2%)	40 inches (±2%)
Speed	T/20	T/20
Depth-of-Field	100 yds to 50,000 yds	2200 yds to 50,000 yds

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3.2.4.11.2 Filters and Shutters. - The lens system shall include six (6) optical quality filters for penetration (color filters), light intensity control, and glare reduction (attenuation of horizontally polarized light) as required for the system to function under the operation conditions specified. The filters shall be arranged as follows:

1. Clear
2. Vertically Polarized
3. Red
4. ND2 + red + vertically polarized
5. ND4 + red + vertically polarized
6. ND8

When (remotely) selected, filters shall be positioned in the optical path of the lens system. The lens system shall also have a sun protector automatic shutter system (high-light sensor and sun shutter) which shall actuate at scene brightness levels of 5×10^4 foot candles with a response time of 80 milliseconds and shall not actuate at 10^4 foot candles. Filters shall operate from the remotely located control panel and shall also function as required in any automatic transmittance control arrangement that may be otherwise specified herein. When filter position is changed the sun shutter shall be positioned in the optical path until the selected filter is in position.

3.2.4.11.3 Reticle Pattern. - The camera shall contain a reticle which will define the limits of the FOV of the long focal length optics when the short focal length optics are in use. The reticle pattern shall appear the same on the monitor when either the long focal length or short focal length lens system is in use. The reticle shall also contain lines which will define the zero look direction of the camera. A typical reticle pattern for the MK 15 MOD 0 Television Camera is shown in Figure 6. The pattern shall be visible on the monitor while the camera is operating at its lowest light level.

3.2.4.11.4 Electronic Reticle Pattern TV Camera Mk 15 MOD 1. - In addition to 3.2.4.11.3 the TV Camera MK 15 MOD 1 shall produce electronic markers which shall consist of horizontal and vertical line segments whose spacing shall correspond to one milliradian in the narrow field of view and 5 milliradians in the wide field of view and two discrete markers. The discrete markers shall indicate when the door apertures are in place and when the lens is in the 5° field of view. A typical electronic reticle pattern is shown in Figure 7.

The milliradian markers shall meet the following requirements:

Location: Vertical plane markers shall be within ±1.0 microseconds of the center of the active horizontal scan time.

Horizontal plane makers shall be within ±2.0 scan lines of the center of the active vertical scan time.

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Stability: Within ±1.0% of the active scan time vertically and horizontally for each marker position.

Length: 20 ± 1 scan lines, horizontal plane markers.

2.9 ± 0.35 microseconds, vertical plane markers.

Spacing: 1 ± 0.1 milliradians in the 1° FOV

5 ± 0.5 milliradians in the 5° FOV

Intensity: Black to white video level internally selectable.

The discrete markers shall meet the following requirements:

Stability: Within ± 2.5% of the active scan time vertically and horizontally for each marker position.

Length: 1.5 ± 0.5 microseconds.

Intensity: Black to white video level internally selectable.

Location: Aperture Marker shall be located in the last 5.5± 2 microseconds of the active horizontal scan time and 2 milliradian markers down from the top of the presentation.

FOV marker shall be located in the last 5.5± 2 microseconds of the active horizontal scan time and 11 milliradian markers down from the top of the presentation.

3.2.5 Monitor Requirements. - Television Monitor MK 30 MOD 0 will, when used as part of Television System MK 5 MOD 0, display video from Television Camera MK 15 MOD 0.

3.2.5.1 Enclosure. - The monitor enclosure shall be drip-proof as defined in MIL-STD-108.

3.2.5.2 Resolution. - Maximum vertical resolution shall be limited only by the use of the 525 line raster. The horizontal resolution shall be at least 900 TV lines in center and 700 TV lines in corners.

3.2.5.3 Input. - The monitor shall operate from an input composite video signal of 1.3 ± .8 volts peak to peak, (black negative), including picture, horizontal and vertical blanking, and synchronizing signals. Monitor input circuits shall be differential and shall have an impedance of 75 ohms.

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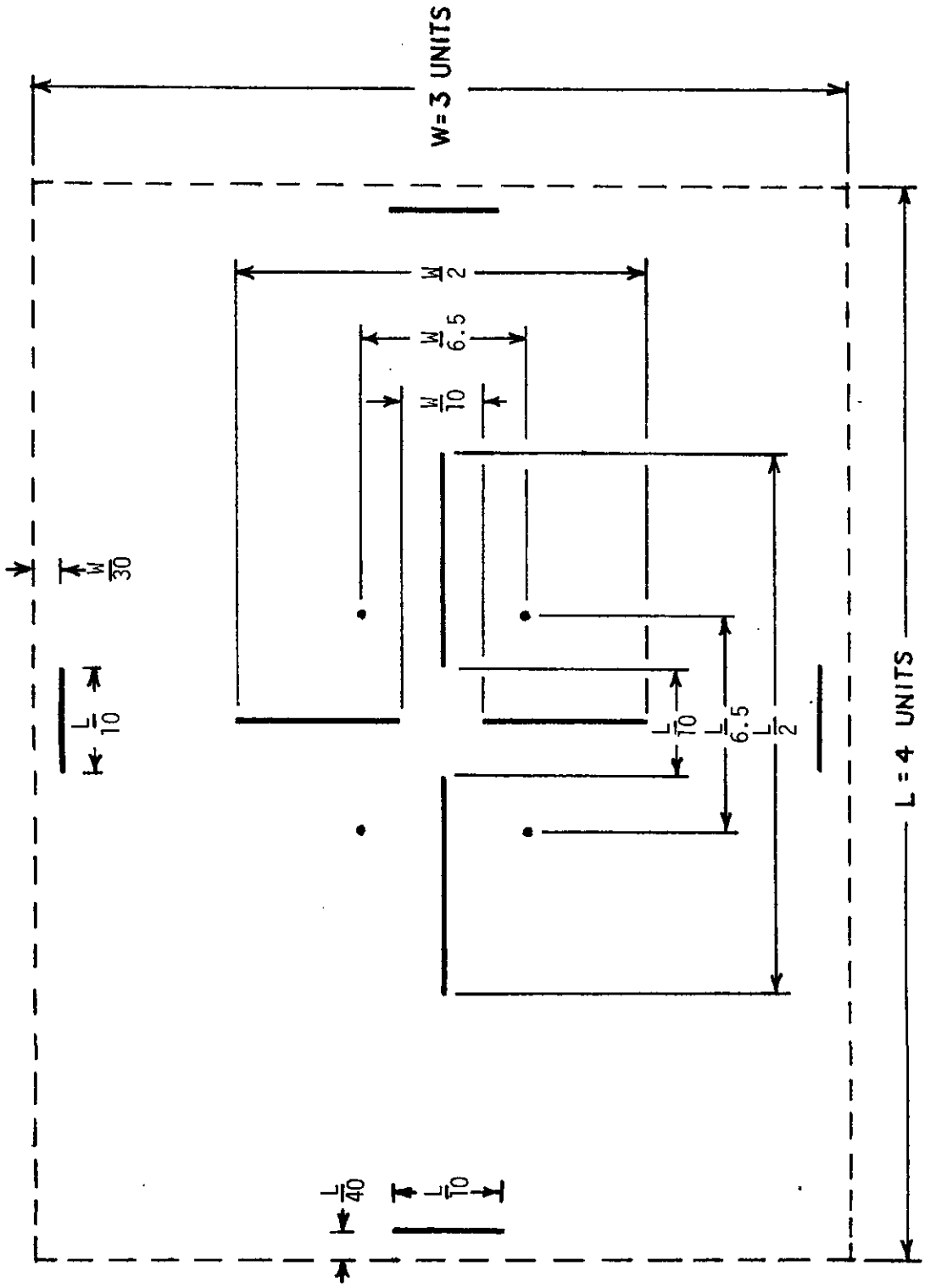


FIGURE 6
TYPICAL RETICLE PATTERN - TV CAMERA MK 15 MOD 0

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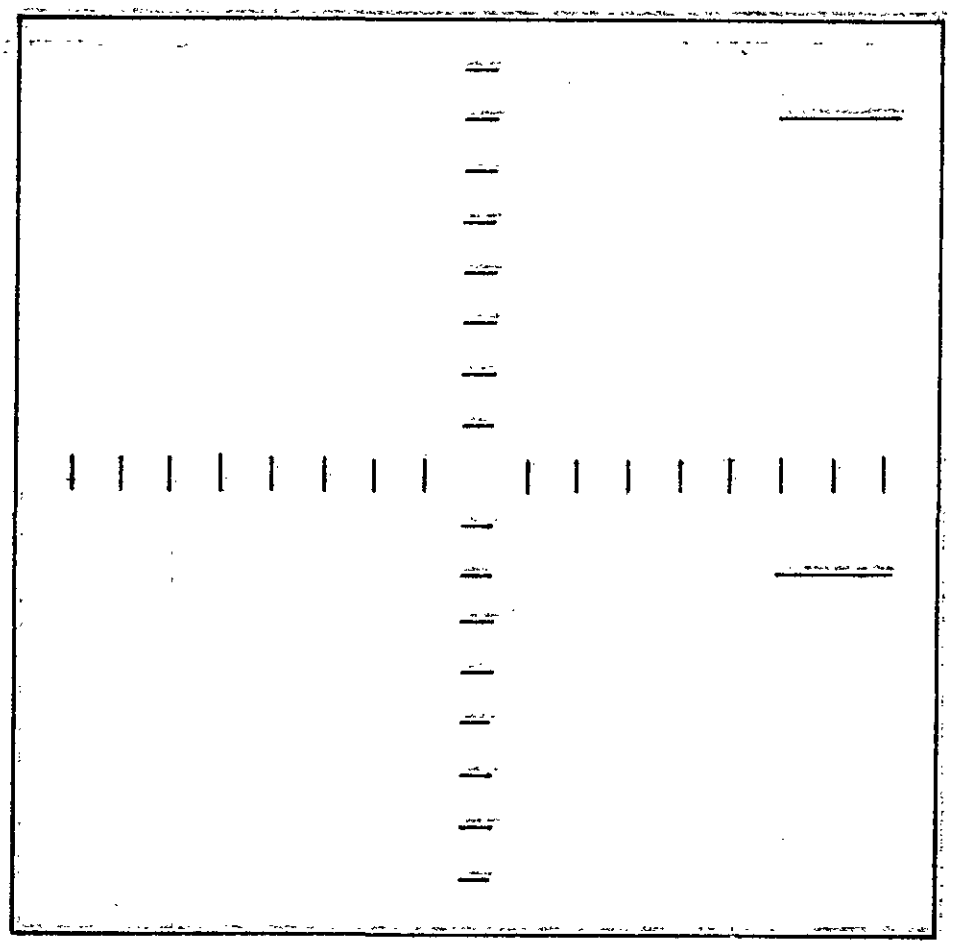


FIGURE 7
ELECTRONIC RETICLE AND MARKERS
TV CAMERA MK 15 MOD 1

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3.2.5.4 Video Bandwidth. - The video preamplifier shall have a bandwidth of 10 MHz flat within ±1 db.

3.2.5.5 Horizontal Scan Rate. - The horizontal scan rate shall be 15,750 Hz with no more than 8.7 microseconds flyback time, and ringing shall cease by 9.7 microseconds.

3.2.5.6 Vertical Scan Rate. - The vertical scan rate shall be 60 Hz.

3.2.5.7 Interlace Ratio. - The interlace ratio shall be 1.84:1, for a frame rate of 30 frames per second. Interlace pairing shall not be present.

3.2.5.8 Picture Aspect Ratio. - The aspect ratio (horizontal to vertical) shall be 4 to 3.

3.2.5.9 Geometric Distortion. - The non-linearity and geometric distortion shall be less than 2 percent of picture height, as measured with a cross hatch or dot pattern.

3.2.5.10 Operational Controls. - Operational controls shall be readily accessible on front panel with respect to the Cathode Ray Tube (CRT) display and shall include power on and off, brightness, and contrast controls. Horizontal and vertical hold operational controls shall also be included and may be located under an accessible cover on the monitor. All other controls shall be considered maintenance controls and may be located in the monitor case.

3.2.5.11 Cathode Ray Tube (Picture Tube). - The monitor shall have a rectangular, 14 inch diagonal picture tube with a P-4 phosphor screen.

3.2.5.12 Display Brightness. - The display brightness of the monitor with bonded safety glass and ENR coating shall be at least 50 foot lamberts without: blooming, loss of resolution and gray scale rendition, increase in distortion, raster shrinkage, and high voltage power supply overloading.

3.2.5.13 Viewing Aids. - The monitor shall be equipped with a hood and bonded safety glass.

3.2.5.14 Airborne and Structureborne Noise. - The monitor shall not generate airborne and structureborne noise in excess of that permitted for Grade C equipment as specified in MIL-STD-740.

3.2.6 Reliability. - The TV System shall have a mean time-between-failure of 500 hours or more at a 95% confidence level.

3.2.7 Maintainability. - The TV System shall have a mean-time-to-repair of 1 hour or less and a maximum-time-to-repair of 2 hours for 95% of all repair actions.

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* 3.3 Construction. - Construction of the LLLTVS MK 5 MOD 0 equipments shall be in accordance with the drawings, documents and specifications set forth in IL.2928410. Construction of LLLTVS MK 5 MOD 1 equipments shall be in accordance with the drawings, documents, and specifications set forth in IL. 2928101-2.

3.3.1 Major Units. - The major equipments of the TV System, where applicable, shall be of fabricated aluminum construction. Subassemblies shall be constructed and mounted so that they are accessible for maintenance purposes. Cable connections to the monitor shall be made by connectors on the back. The camera connector shall be located on the side.

3.3.1.1 On Mount Equipment. - The Camera MK 15 MOD 0 mounted on the rotating member of the director structure, shall withstand without damage an elevation speed of .3 radians per second (rad/sec), and a maximum angular deceleration of 45 radians per second per second (rad/sec/sec). Similarly, the Camera (offset from director rotational center by 4.3 ft. ± 1 in. in train and 2.8 ft. ± 1 in. in elevation) shall withstand without damage a train speed of 3 rad/sec maximum, and a maximum normal angular acceleration of 4.5 rad/sec/sec and a maximum braking deceleration of 10 rad/sec/sec. No damage shall be caused to any of the on-mount equipment as a result of these velocities and accelerations, together with any combination of roll pitch and yaw as specified in 3.5.9 and a maximum ice loading as as specified in 3.5.4 (except that the combination of ice load and 45 rad/sec/sec deceleration is not required). The Camera MK 15 MOD 1 shall withstand without damage the acceleration and braking forces of the Optical Sighting System MK 1 MOD 2.

3.4 Physical Characteristics. - The TV System shall have physical characteristics as follows:

3.4.1 Weight. - The total system weight shall not exceed 280 pounds. Individual equipment weights shall not exceed the following maximums:

a. Camera, TV MK 15 MOD 0 and MOD 1 - 170 lbs.
b. Monitor, TV MK 30 MOD 0 - 90 lbs.

3.4.2 Size. - The TV System equipment shall not exceed the following maximum dimensions:

a. Camera, TV MK 15 MOD 0 and MOD 1 - 55" L x 15" W x 20" H
b. Monitor, TV MK 30 MOD 0 - 20" L x 16" W x 13" H

3.4.3 Mounting Requirements

* 3.4.3.1 Camera. - The Camera, MK 15 MOD 0, shall be compatible with the requirements of NAVORD Dwg. No. 2807052. The Camera MK 15 MOD 1 shall be compatible with the requirements of NAVORD Dwg. No. 2928507.

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3.4.3.2 Monitor. - The Monitor shall be base mounted by studs bolted to a platform provided by the installing activity. Mounting hole pattern will be as shown in Figure 8.

3.4.3.3 Controls MK 5 MOD 0. - TV System controls installed in Radar Set AN/SPG-51C/D consoles will be accomplished as directed by the Naval Sea Systems Command. TV System Controls installed in Radar Set AN/SPG-51D/C consoles will be accomplished by a separate ORDALT.

3.4.3.4 Controls MK 5 MOD 1 Television System. - TV System controls installed in the Gun Control Console MK 68 MOD 4 and covered herein by paragraph 3.1.5.2.2 shall control the MK 15 MOD 1 camera.

3.5 Environmental Requirements. - TV System equipments shall operate as specified herein under any of the following conditions or combinations thereof.

3.5.1 Temperature. - The ambient temperature limits for the camera equipment located on the director and exposed to the elements shall be -28°C and 55°C. The ambient temperature limits for the Monitor located in a protected area shall be 0°C to 50°C. In addition, the equipment shall be suitable for storage at any ambient temperature from -40°C to 60°C without damage. A 25% decrease in resolution is acceptable at ambient temperature above 50°C.

3.5.2 Humidity. - The TV System shall be capable of starting and operating in atmospheres having a relative humidity as great as 95% in temperatures up to 40°C and a constant water content (rain type conditions) at temperatures of 40°C and above.

3.5.3 Wind. - The camera equipment exposed to the elements shall operate normally in winds having a relative velocity of 75 knots and shall not be damaged when subjected to winds having a relative velocity of 100 knots when the director is in the stowed position.

3.5.4 Icing. - Provided the camera heater has been energized the camera equipment shall start and operate when subjected to conditions which would produce a maximum ice load of 4.5 pounds per square foot. At points of limited clearance (1/8" or less), this requirement shall apply after the equipment has been manually broken free of ice.

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3.5.5 Salt Atmosphere. - The exposed equipment shall be capable of satisfactory operation after the equipment has been exposed to the salt spray test described in MIL-F-18870.

3.5.6 Solar Radiation. - The TV camera group will be protected by a sun shield, however, the exposed equipment shall operate normally when exposed to direct solar radiation as defined in MIL-STD-210.

3.5.7 Vibration. - The equipment shall be capable of withstanding the vibration requirements of MIL-STD-167 Type I. In addition, the on mount equipment shall withstand frequencies from 10 to 125 Hz at 1 Hz discrete intervals for 1 minute at each discrete frequency and a magnitude of 0.050 inch double magnitude, or 1.3 g's peak acceleration, whichever is less. In conjunction with this, a 600 Hz bandwidth noise centered at 700 Hz of 0.003 g's per Hz acceleration density shall be superimposed on the discrete frequency for exposed equipment.

3.5.8 Shock. - Units of the TV System shall be subjected to shock loading as set forth in MIL-S-901 for Grade B, subsidiary components, Class II for exposed equipment and Grade B, deck mounted, Class II for protected equipment except that shock force shall be limited to 75 g's.

3.5.8.1 Gun Blast Shock. - The TV System shall operate as specified herein without readjustment of internal controls, after being subjected to gun blast shock loadings corresponding to 15 g's at 200 Hz for a period of 40 milliseconds.

3.5.9 Ship's Motion. - The equipment shall be capable of normal operation under the following conditions of ship's motion and director motion specified in 3.3.1.1

a. Maximum roll amplitude of ±30° within a period of 9 seconds.
b. Maximum pitch amplitude of ±5° within a period of 6 seconds.
c. Maximum yaw amplitude of ±2.5° within a period of 6 seconds.

In addition, the equipment shall not be damaged with director in the stowed position under the following conditions of ship's motion:

a. Maximum roll amplitude of ±45° within a period of 9 seconds.
b. Maximum pitch amplitude of ±15° within a period of 6 seconds.
c. Maximum yaw amplitude of ±7.5° within a period of 6 seconds.

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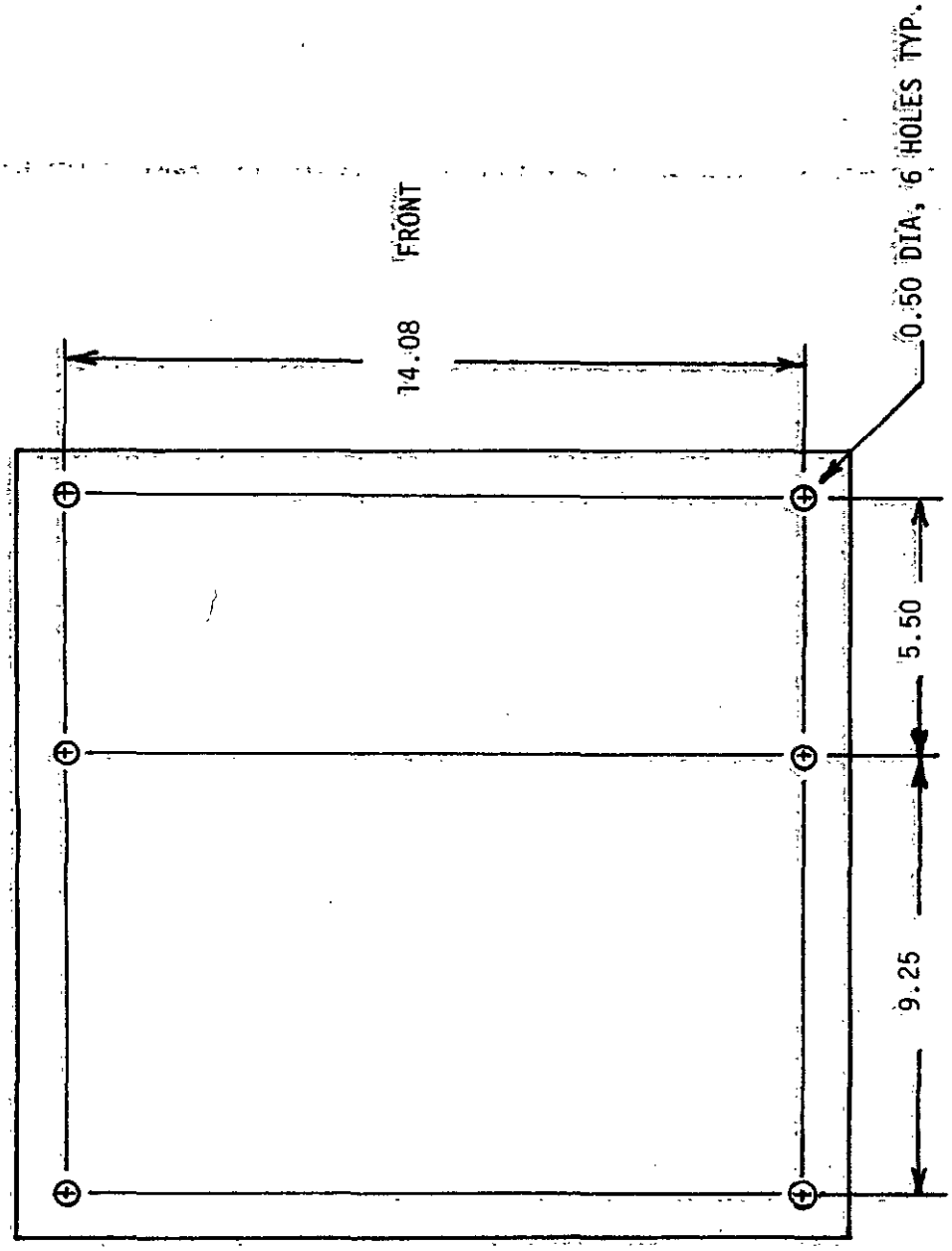


FIGURE 8
HOLE PATTERN
FOR MOUNTING MONITOR

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3.5.10 Inclination. - Tests shall be conducted in accordance with the requirements of MIL-F-18870. Inclination tests shall be conducted with the equipment operating.

3.5.11 Water Tightness. - The TV camera MK 15 shall be water tight when operated in each or any combination of each environment in 3.5.1 through 3.5.7.

3.6 Maintenance Kit Carrying Case. - The carrying case for the maintenance kit shall meet the following environmental requirements.

3.6.1 Vibration. - The carrying case shall meet vibration requirements as specified in MIL-T-28800.

3.6.2 Temperature. - The carrying case shall have storage temperature limits of -40° to 71°C.

3.6.3 Humidity. - The carrying case shall meet humidity requirements as specified in 3.5.2.

3.6.4 Shock. - The carrying case shall meet shock requirements as specified in MIL-T-28800.

3.6.5 Salt Atmosphere. - The carrying case shall meet salt atmosphere requirements as specified in 3.5.5.

3.7 Electromagnetic Interference Control (EMI). - EMI for the TV System shall be as set forth in Table II, Equipment Class IC of MIL-STD-461. The following test requirements shall not be applicable: CE01 (DC only); CE03, Notice 3 (AC only); CE05 (optional); CS03; CS04; CS05; (T)CS07; (T)RE04; RS01; RS02; and T(RS04).

3.8 Marking. - Marking of equipment components, cabling, wires, etc., shall be in accordance with MIL-F-18870 and as set forth on the drawings listed on IL 2928400.

3.9 Identification Plates. - Identification plates for equipment, units and assemblies shall be fabricated and marked in accordance with MIL-STD-130.

3.10 Workmanship. - Workmanship shall be in accordance with Requirement 9 of MIL-STD-454.

3.11 Interchangeability. - All parts shall be interchangeable physically and electrically with corresponding parts without drilling, filing, or the use of undue force. The replacement part when installed shall not result in a departure from the performance specified herein.

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3.12 Safety. - The equipment shall provide maximum protection for personnel from electrical, mechanical, and health hazards in accordance with MIL-STD-454, Requirement 1.

3.13 Human Engineering. - Human engineering design criteria shall be as specified in MIL-H-46855.

3.14 Documentation. - All documentation shall be as specified in the contract or order.

3.15 Logistics

3.15.1 Maintenance. - Provisions for maintenance of the director mounted camera shall be limited to changing the printed boards in the camera electronics and replacement of actuators in the optical subsystem. If further correctional maintenance is required construction of the camera shall be such that the camera electronics and the camera head assembly can be disassembled from the optics assembly and taken below deck for trouble shooting. For maintenance purposes the following items shall be considered replaceable aboard ship below deck:

a. Camera head (complete assembly)

b. Printed boards (camera and monitor)

c. Primary power supply components (transformers, chokes, capacitors, EMI filters) (camera and monitor)

d. Secondary power supply (camera)

e. Printed board components (camera and monitor) (in emergencies only)

f. Motors in optics assembly

3.15.1.1 Accessiblility. - All printed circuit boards shall be readily accessible for replacement or repair. Cable entrances shall not be made in the front of the equipment. Each subassembly capable of being completely removed from its associated housing shall be provided with conveniently located terminal boards or connectors to permit all interconnecting cables to be readily disconnected. If plugs and jacks are used, means shall be provided for their correct alignment. Controls shall not be located in covers that must be removed for servicing of the equipment.

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3.15.1.2 Test Points. - Test points shall be provided in readily accessible locations for measuring, observing, or injecting those quantities needed for maintenance and repair of the equipments.

3.15.1.3 Test Connectors. - Connectors, jacks, and plugs used as test points shall be properly mated with standardized test equipments.

3.15.1.4 Spare Terminals and Pins. - Terminal Boards and multiple connectors used with interconnecting major units and assemblies shall contain spare terminal and spare connector pins in the amount of at least 10% of each, and in no case less than two spare terminals and spare pins. The 10% spare terminals or pin allowance may be concurrent with the spare lead requirements of cabling.

3.15.1.5 Convenience Outlets. - No convenience outlets shall be provided in the equipment.

3.15.1.6 Built in Test Facilities. - Built-in test facilities shall be provided whenever any of the following conditions apply:

a. When periodic observations are necessary, each time the equipment is turned on or more frequently if required.

b. When the use of portable equipment will not give consistent reproducible results for operations necessary in maintenance.

c. When disassembly of an equipment is necessary to measure a quantity required for maintenance of the equipment.

d. When the maintenance of complex equipment can be facilitated by fault isolation test devices.

e. When measurements which are dangerous to life must be made.

3.15.1.7 Portable Test Sets and Tools

3.15.1.7.1 Portable Test Sets. - Portable test sets are defined as those test equipments which are either hand portable or can be rolled or moved from one location to another without difficulty. Test sets are classified as "test set common" or "test set Peculiar". Test sets common are considered to be standard items in the naval establishment and aboard ship and are used for two or more equipments. Test sets peculiar are sets that are unique to one equipment only.

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3.15.1.7.2 Tools. - Special tools for the TV System shall be as specified herein or as otherwise set forth in the contract or order.

3.15.1.8 Test Cables. - The test cables assemblies as provided in the MK 86 MOD 0 and MOD 1 maintenance kit are identified and defined as follows:

a. Cable Assembly Dwg. No. 2928425-1, power/control cable for the camera.

b. Cable Assembly Dwg. No. 2928076-1, one (1) signal (video) cable with adapters for the monitor.

c. Cable Assembly Dwg. No. 2928469-1, power cable extension for the monitor.

d. The power/control cable shall be provided with connectors that will match MS17346R028-C12P and MS3102E-24-285 type connectors. The signal video cables shall be provided with type N connectors. Cables shall have a maximum length of 15 feet.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality Conformance Inspections. - Each system offered for acceptance under contract shall be subjected to the examinations and tests specified in 4.2.1 through 4.2.5 and 4.2.13. In addition, the sample selected from units having satisfactorily passed these inspections shall be subjected to the tests of 4.2.6, 4.2.7, 4.2.8, 4.2.11, and 4.2.12 at an activity designated by the contracting officer. Sampling for the tests of 4.2.6, 4.2.7, 4.2.8, 4.2.11,

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and 4.2.12 shall be as determined by the contract or order. Sampling for verification of reliability and maintainability as specified in 4.2.9 and 4.2.10 shall be as determined by the contract or order. Failure to meet any of the inspection requirements shall be considered cause for rejection.

4.2.1 Examinations. - The system shall be subjected to examinations to determine conformance with requirements set forth herein including:

a. Parts, materials, and processes
b. Cabling and wiring
c. Accessibility
d. Finish and color
e. Identification plates
f. Test facilities (test points)
g. Operating controls and indicators
h. Workmanship
i. Marking
j. Packaging and packing

(Examinations are those inspections which can be made by visual, tactile, olfactory, or mechanical means to verify that requirements have been met.)

4.2.2 Performance Tests. - The system shall be subjected to tests to determine conformance with requirements set forth herein. The tests shall be arranged to correlate with the specific performance requirements set forth herein including tests for:

a. Resolution
b. Video content and level
c. Optical performance

(Tests are those inspections which can be made by applying predetermined stimuli to the item under test and observing, measuring, or comparing evoked responses to ascertain that a desired performance has been obtained.)

4.2.3 Inspection Conditions

4.2.3.1 Standard Conditions. - Unless otherwise specified all inspections shall be made under the following nominal conditions:

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a. Room ambient temperature	25 degrees C
b. Normal atmospheric pressure	Approximately 29.92 inches Hg
c. Room ambient relative humidity	50 percent

4.2.3.2 Attitude. - Unless otherwise specified the equipment shall be tested in its normal operating position.

4.2.3.3 Measurements. - During all tests, the accuracy of instruments and test equipment used to control or monitor test parameters shall be verified periodically to the satisfaction of the procuring activity. All instruments and test equipment used in conducting tests specified herein shall:

a. Conform to laboratory standards where calibration is traceable to the prime standards at the US Bureau of Standards.

b. Have an accuracy of at least one-fifth tolerance for the variable to be measured.

4.2.4 Packaging and Packing Inspection. - Inspection shall be conducted in accordance with MIL-E-17555 to assure compliance with the requirements of 5.1 and 5.2 herein.

4.2.5 Marking Inspection. - Examinations shall be conducted to ascertain that units, maintenance parts, and other items prepared for shipment are marked in accordance with the requirements of 5.3 herein.

4.2.6 Environmental Tests

4.2.6.1 Nonoperating Temperature Test. - The system shall be subjected to the nonoperating temperature test specified in MIL-F-18870 except that the maximum temperature shall be 67 ± 3 degrees C.

4.2.6.2 Operating Temperature Test (Class 2). - The camera shall be subjected to the operating temperature tests for Class 2 equipment as specified by MIL-F-18870 except that maximum ambient temperature shall be 55 ± 2 degrees C. The monitor shall be subjected to the operating temperature test for Class 4 equipment as specified by MIL-F-18870

4.2.6.3 Humidity. - The TV System shall be tested for resistance to humidity in accordance with MIL-F-18870 to determine that the systems meet requirements specified herein except that the water content shall be as specified in 3.5.2 herein and the maximum temperature as specified in 3.5.1 herein.

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4.2.6.4 Inclination Test. - The camera shall be subjected to the inclination test set forth in MIL-F-18870.

4.2.6.5 Vibration Test. - The camera shall withstand vibration tests for Type I equipment in accordance with MIL-STD-167. In addition, the equipment shall withstand frequencies from 10 to 125 Hz and 1 Hz discrete intervals for 1 minute at each discrete frequency and a magnitude of 0.050 inches double magnitude, or 1.3 g's peak acceleration whichever is less. In conjunction with this, a 600 Hz bandwidth noise centered at 700 Hz of 0.003 g's per Hz acceleration density shall be superimposed on the discrete frequency. The monitor shall withstand vibration tests for Type I equipment in accordance with MIL-STD-167.

4.2.6.6 Shock Test. - The camera and monitor shall be subjected to shock tests for Grade B, Class II; lightweight, Type B equipment in accordance with MIL-S-901, except as specified in 3.5.8 herein.

4.2.6.7 Degree of Enclosure

4.2.6.7.1 Lens Assembly and Sensor Enclosure. - An airtight test shall be conducted on the Lens Assembly and Sensor Enclosures in accordance with MIL-STD-108. Equipment shall be non-operating.

4.2.6.7.2 Camera Electronics Enclosure. - A watertight test shall be conducted on the camera enclosure in accordance with MIL-STD-108. Equipment shall be non-operating.

4.2.6.7.3 Monitor Enclosure. - A drip-proof enclosure test shall be conducted on the monitor in accordance with MIL-STD-108. The equipment shall be nonoperating.

4.2.6.8 Salt Spray Tests. - A salt spray test in accordance with FED-STD-151 shall be conducted on all parts with coatings or finishes, and to frames and enclosure structures assembled for final use. The test shall not be conducted on complete sets of major units. Sample corner structures and other critical sections shall be used for the test. Duration of the tests shall be 200 hours for frame and enclosure structures and 48 hours for individual parts.

4.2.7 EMI Tests. - The system shall be tested for radio frequency and low frequency noise interference in accordance with the methods set forth in MIL-STD-462 for Equipment Class IC. Test Requirements CE01 (DC only); CE03, Notice 3 (AC only); CE05 (optional); CS03; CS04; CS05; (T) CS07; (T) RE04; RS01; RS02; and T (RS04) are not applicable. The maximum radiated and the maximum conducted noise interference and electrical disturbance shall not exceed the values specified therein.

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4.2.8 Power System Variations. - The system shall be tested for satisfactory operation under the power system variations characterizing Type II power specified herein. This test may be conducted during the performance of any or all of the tests of this specification at the discretion of the testing activity or as specified by the contract or order.

4.2.9 Reliability. - Reliability requirements as specified in 3.2.6 herein shall be demonstrated in accordance with paragraph 5.3 of MIL-STD-785.

4.2.10 Maintainability. - Maintainability requirements as specified in 3.2.7 herein shall be demonstrated in accordance with paragraph 5.11 of MIL-STD-470.

4.2.11 Airborne and Structureborne Noise. - The monitor shall be tested for airborne and structureborne noise in accordance with the methods set forth in MIL-STD-470.

4.2.12 Gun Blast Shock Test. - The TV Camera shall be subjected to gun blast shock tests of 15 g's at 200 Hz for 40 milliseconds. The test shall be repeated 10 times at 2 second intervals. The TV Camera shall operate as specified herein after the test without requiring adjustment of internal controls.

4.2.13 Water Tightness. - The TV Camera MK 15 shall be subjected to a leak test whereby a vacuum of minus(-) 3 psig (+10% -0%) is applied. The unit will be considered acceptable if the vacuum is maintained within 0.5 psig/hour. The test duration will be a minimum of one (1) continuous hour.

5. PACKAGING

5.1 Preservation - Packaging. - Preservation - packaging of the TV systems shall be in accordance with MIL-E-17555 level A or C, as specified (see 6.2).

5.2 Packing. - The TV system shall be packed in accordance with MIL-E-17555 level A or C, as specified. (see 6.2).

5.3 Marking. - Shipping containers shall be marked in accordance with MIL-STD-129, including serial numbers.

6. NOTES AND CONCLUDING MATERIAL

6.1 Intended Use. - The equipment specified herein, when operating as the Television System MK 5 MOD 0, is intended to provide a method of angle tracking for Fire Control System MK 74 when satisfactory track cannot otherwise be maintained and the television system

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MK 5 MOD 1, is intended to provide optical sighting information for the gun fire control system MK 86.

6.2 Ordering Data. - Procurement documents should specify the following:

6.2.1 Procurement Requirements

a. Title, identification number and date of this specification.

b. Documentation required, if any (see 3.3).

c. Sampling plan for cameras to be subjected to the inspections and tests of 4.2.6, 4.2.7, 4.2.8, 4.2.11, and 4.2.12 (see 4.2).

d. Verification of reliability requirements (see 4.2.9).

e. Verification of maintainability requirements (see 4.2.10).

f. Level of packaging and packing (see 5.1 and 5.2)

g. Power system variation test requirements (see 4.2.8).

6.2.2 Contract Data Requirements. - Any data required for delivery in connection with this document must be specified in a contract clause or on a DD Form 1423 incorporated in the contract. If test reports are included in this data they should conform to MIL-STD-831.

6.2.3 The attention of the contracting officer is invited to the Quality Assurance Provisions and options in MIL-P-116.

6.2.4 Criteria for the use of proper level of preservation, packaging; and packing shall be as follows:

For Level A - This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

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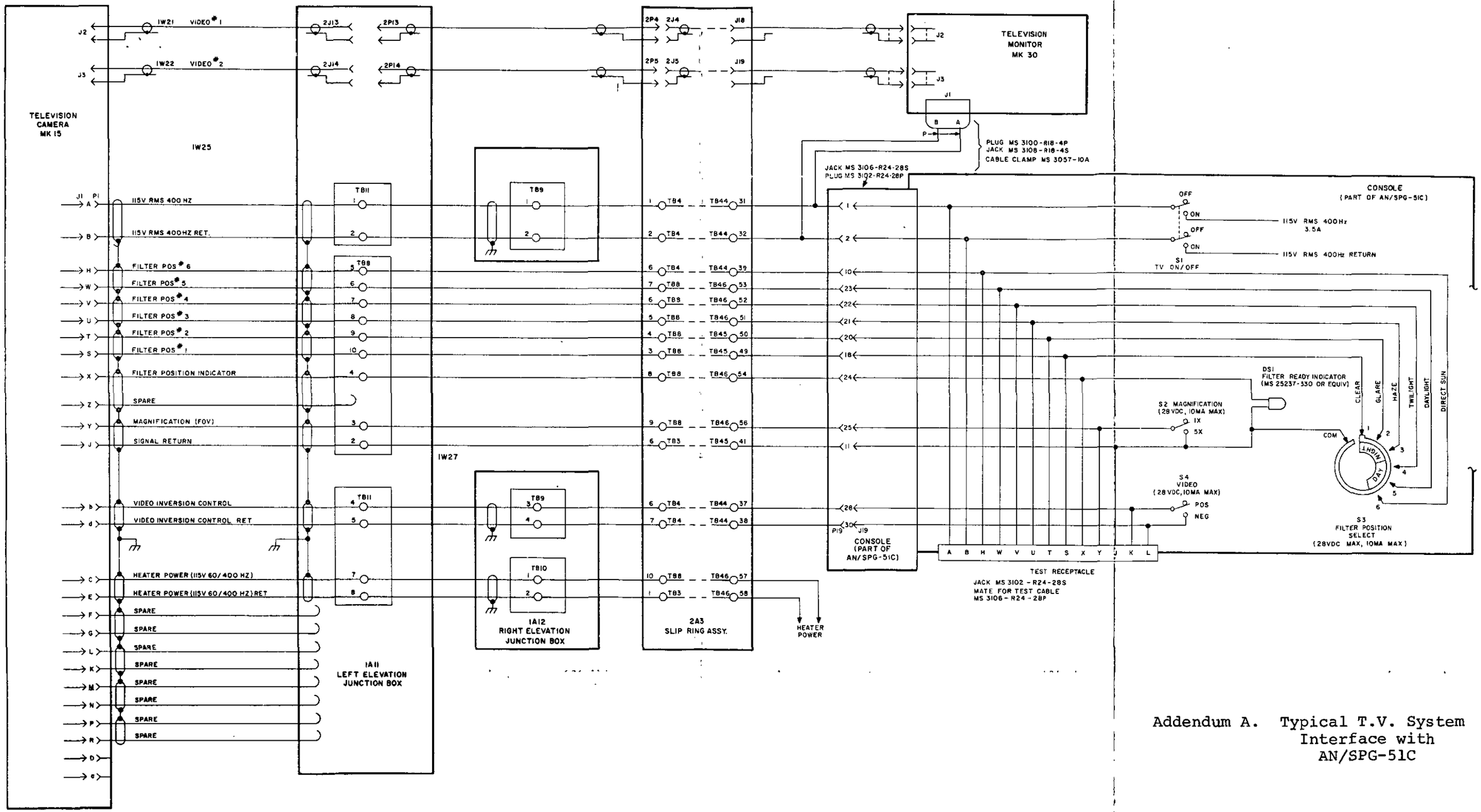
For Level C - This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

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NAVY-OS

Preparing Activity
Navy - OS
Project Number
1290-N329

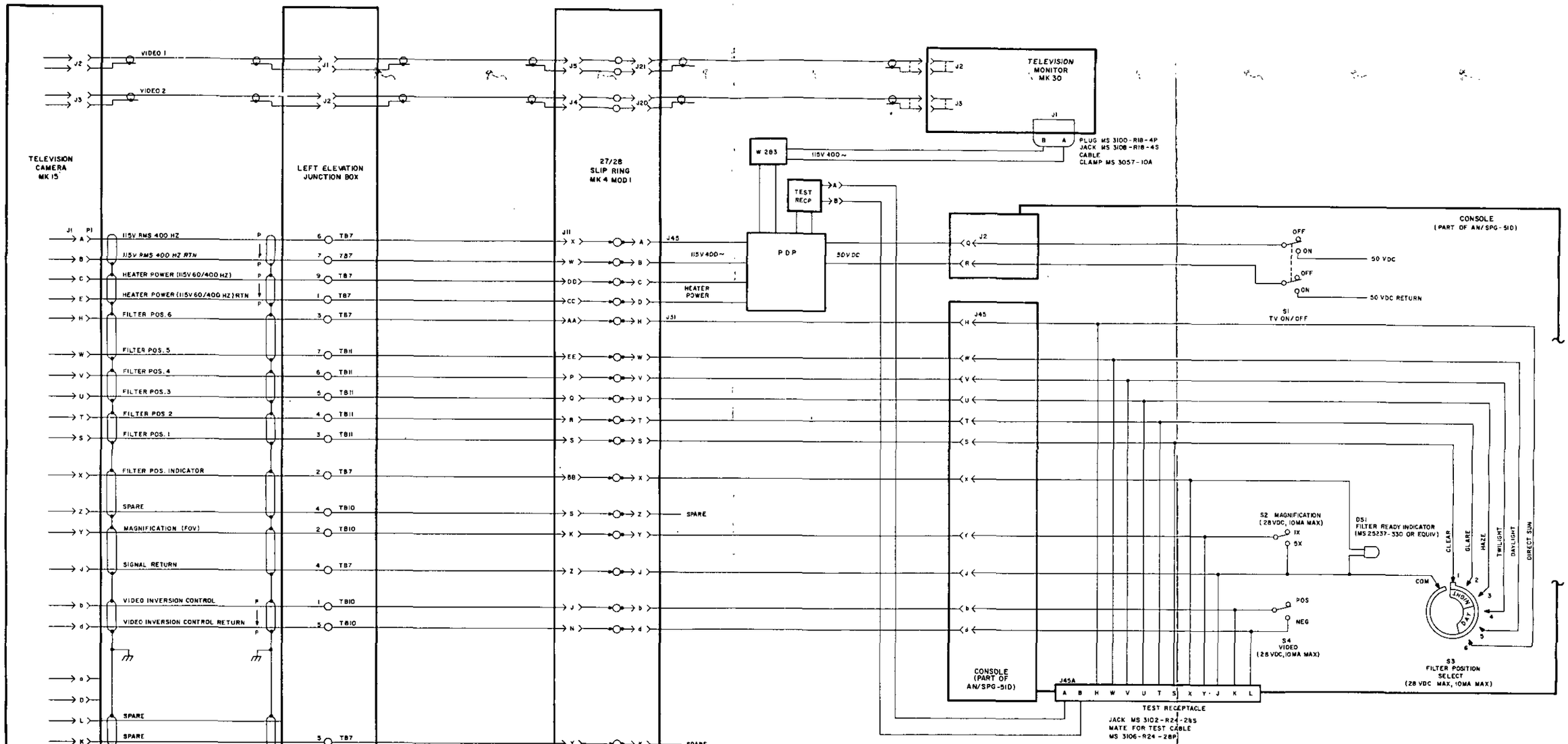
NOTE: The margins of this revision have been marked with an asterisk (*) to indicate those paragraphs containing significant changes from the previous revision. Minor changes are too numerous to indicate.

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Addendum A. Typical T.V. System Interface with AN/SPG-51C

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Addendum B. Typical T.V. System Interface with AN/SPG-51D

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DOCUMENT IDENTIFIER AND TITLE

MIL-T-82607B (OS) TELEVISION SYSTEM MK 5 MODS 0 and 1

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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